

1. Moving Signals

Solutions to equations of motion in physics, such as the wave equation, often look something like

$$y = f(x - ct) \tag{1}$$

If you aren't used to looking at functions like this, it can be a bit hard to get an intuitive sense for what that function is telling us and how to use it.

2. Simplest Lorentzian

Let's start by looking at a concrete function example,

$$z = \frac{1}{1 + x^2} \tag{2}$$

The smallest value x^2 can ever have is zero, so the smallest value $1 + x^2$ can ever have is 1. That happens when $x = 0$. For large positive and negative values of x , then $1 + x^2$ is even larger and positive. When we take $\frac{1}{1+x^2}$, then the largest value we get is 1 divided by the smallest value, which we saw is 1. For large positive/negative values of x , then $\frac{1}{1+x^2}$ will be very small. We can combine these properties to see that $\frac{1}{1+x^2}$ will have a bump of height 1 centered at $x = 0$, and falling away from there. If we plot it, we indeed see that is what our function looks like, as shown in Figure 1. Incidentally, this function goes by many names, and in physics is most commonly known as a Lorentzian. It shows up in several areas of physics, but we're using it here just as the simplest mathematical function that has a peak and goes to zero away from the peak.

3. Shifted Lorentzian

Let's make our function a little more complicated now. We'll add an extra parameter x_0 , and look at the function

$$z = \frac{1}{1 + (x - x_0)^2}$$

When we use this function, we'll pick a value for x_0 and hold it constant. We'll then look at how z depends on x after we've fixed the value of x_0 . The function looks very similar to our previous one, except for the x_0 in the denominator. We can do the same analysis as before, where we start with the behaviour of $(x - x_0)^2$. Since we have a number squared, the smallest value it can ever have is zero, and it will get large as we move away in either direction from that zero. The minimum of $(x - x_0)^2$ will happen when we're squaring zero, so we want to find where

$$x - x_0 = 0 \tag{3}$$

$$x = x_0 \tag{4}$$

When $x = x_0$, our squared value will be zero, and it will be larger otherwise. If we take that fact and go through the exact same reasoning process as we did for $z = \frac{1}{1+x^2}$, then we expect we'll have a peak where $x = x_0$ and dropping away from that in either direction, and that our peak function

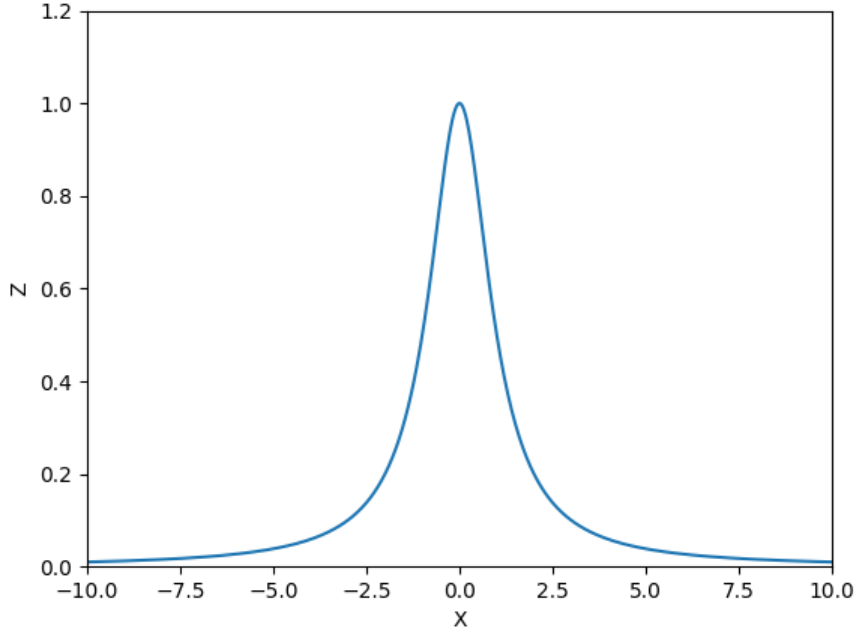


Fig. 1.— Graph of the function $z = \frac{1}{1+x^2}$. As expected, it peaks at $x = 0$ with a value of 1, and it decays towards zero as x gets large in either the negative or positive directions.

value will be 1. In short, we're going to get the exact same function, except we'll scoot it over so that the center of the peak is at $x = x_0$ instead of $x = 0$. Figure 2 shows our function now for a few different values of x_0 . As you can see, the *shape* of the function does not change, but we are sliding it left and right as we change x_0 .

4. Time-Varying Lorentzian

We now have one final generalization to make. We're going to make the value of x_0 depend on time t , so that

$$x_0 = vt \quad (5)$$

For reasons that will become clear, the constant value v is the *velocity*. Our function now becomes

$$z = \frac{1}{1 + (x - vt)^2} \quad (6)$$

We'll start by looking at this for selected, fixed values of time. If we pick a time, then our function will peak at $x = vt$. If we set $v = 1$, then at time $t = 0$, our peak is at $x = 0$. At time $t = 1$, the peak is at $x = 1$, at time $t = 2$ the peak is at $x = 2$ and so on. In other words, as time increases, our function moves to the right. If we set $v = 2$, at $t = 0$ the peak is still at $x = 0$, but at $t = 1$, then the peak is at $x = 2$, and $t = 2$ the peak is at $x = 4$ *etc.* The function is still moving to the right, but it is going *faster* since the location of the peak moves further to the right in the same

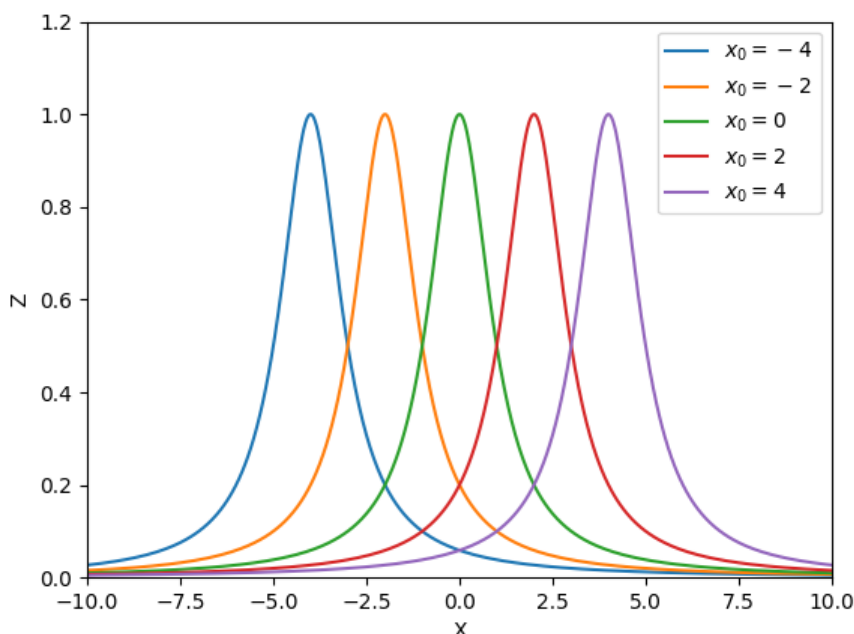


Fig. 2.— Graph of the function $z = \frac{1}{1+(x-x_0)^2}$ for $x_0 = [-4, -2, 0, 2, 4]$. Each curve corresponds to a separate value of x_0 , and we hold that value of x_0 fixed for that curve. Now each curve peaks at $x = x_0$ instead of $x = 0$ as in Figure 1. The overall shape of the function does not change as we change x_0 , but as we change x_0 , we are sliding the function left and right without changing its shape.

amount of time. If we set $v = -1$, then at $t = 0$ we still peak at $x = 0$, but now at $t = 1$ the peak is at $x = -1$, and at $t = 2$ the peak is at $x = -2$. The function is now moving to the *left*. The *sign* of v tells us the direction the peak moves, and the *magnitude* of v tells us how quickly it is moving. That’s exactly the velocity.

Rather than trying to insert an animation into a PDF, I suggest you go to [desmos.com](https://www.desmos.com). Desmos will animate functions for you so you can see what’s happening. If you want to play with different velocities, click on the “graphing calculator” box, and then I suggest you enter in box 1 “ $v=1$ ” and hit enter. That will bring up box 2, where you should enter $1/(1+(x-vt)^2)$. Then you can add a slider to make an animation as t changes. I suggest you set the slider animation properties to be have a loop with time increasing – this will be two arrows pointed to the right (the default is to have time increase then decrease, which can be a bit confusing). As you increase v , you’ll see the velocity increase, and you’ll see that if v is positive/negative, the peak moves to the right/left.

We could also pick a value for x , and look at the height there as a function of time. An analogy would be a swimmer out floating in the ocean as a wave passes by. The swimmer starts out at sea level, but as the wave passes by, the swimmer goes up, then goes back down. If we plot the swimmer’s height vs. time, we’ll see a peak centered at the time the wave goes by the swimmer’s location. Visually, the plot will look exactly like the ones we’ve already seen, only this time we’re

plotting height against time at a fixed position, instead of height against position at a fixed time.

More generally, we have z as a function of both position x and time t . To visualize this, we could make an image where x and t are the axes, and the color of the image at a given pixel corresponds to the function value at that position and time. This 2-d plot is in Figure 3 with $v = 1$. Time increases from bottom to top, so the figure shows that the peak moves to the right as time increases. If we want to see the full curve at a given point in time, we take the horizontal slice at that time. If we want the height vs. time at a position, we take the vertical slice at that position. I've also make the 2-d plot with $v = -1$ in Figure 4, where you can see the peak moves to the left as time increases.

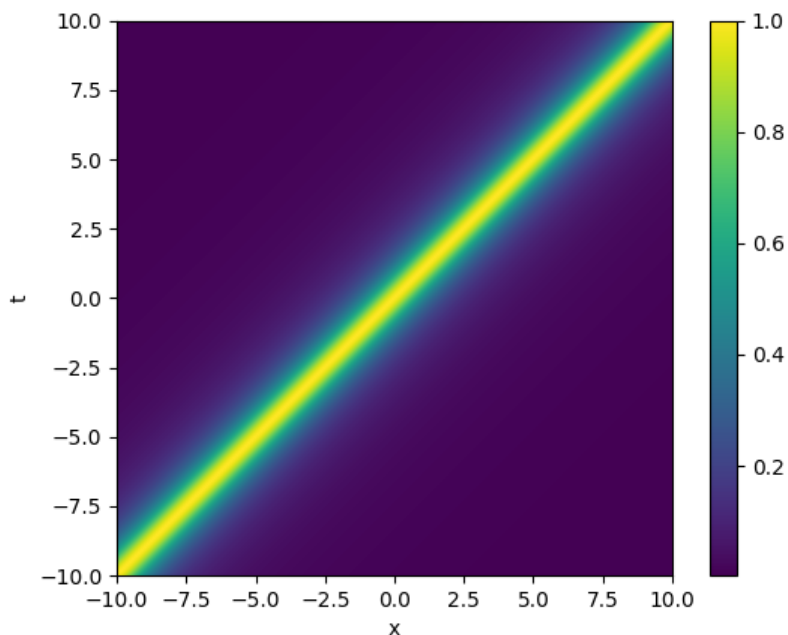


Fig. 3.— Graph of the function $z = \frac{1}{1+(x-t)^2}$, portrayed as a 2-d array. To see our function everywhere in space at a given time, we find that time on the y -axis and take the horizontal slice from the 2-d array. Similarly, if we want to find the height as a function of time at a certain location, we find our desired location on the x -axis, and show the vertical slice that goes through that location. The fact that the band moves to the right as time increases (goes up) tells us the velocity is positive.

5. Sin/Cos and Wavevector

You might notice that if we take $f(x - ct)$, that the argument to our function has units (in this case, metres if x is in metres). If we wanted to use a sin or cosine instead, the inputs to sin/cos are radians, which don't have any units. In this case, the usual convention is to write

$$y = f(kx - \omega t) \tag{7}$$

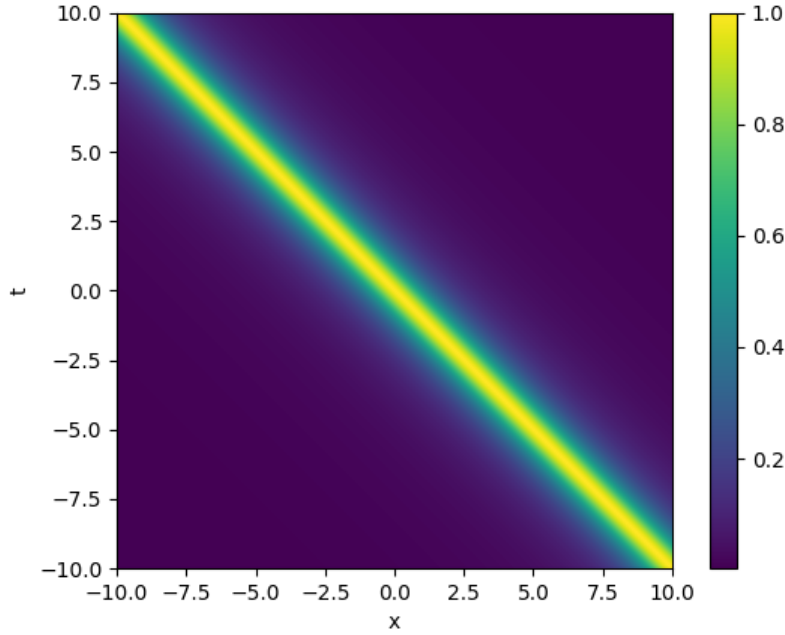


Fig. 4.— Graph of the function $z = \frac{1}{1+(x+t)^2}$, portrayed as a 2-d array. By changing the sign, we have set $v = -1$, and so we expect our peak to move to the left as time increases. Qualitatively, everything looks the same as in Figure 3, except by changing the sign of v , the peak moves to the left instead of the right.

In this case, our new function f could be a sine or a cosine, or it could really be anything that takes a dimensionless quantity as an input. What happens now?

First, look at the units. If x is in metres, and t is in seconds, then k must have units of m^{-1} and ω must have units of s^{-1} (better known as Hz!). That makes both kx and ωt dimensionless, so we can feed them into our function f that is expecting a dimensionless input. Now we can take Equation 7 and pull out a factor of k to get:

$$y = f\left(k\left(x - \frac{\omega}{k}t\right)\right) \quad (8)$$

We can define a new function g such that $f(kx) = g(x)$. That leaves Equation 8 like:

$$y = g\left(x - \frac{\omega}{k}t\right) \quad (9)$$

This looks a whole lot like the function we started with, as long as we set

$$v = \frac{\omega}{k} \quad (10)$$

We'll see this a whole lot over the course. We usually call k the wave vector (more-or-less $1/\text{wave-length}$) and ω is the frequency. For sine/cosine, then ω is the *angular* frequency, *i.e.* how long it take to change the input to sine/cosine by 1 radian. An entire cycle takes 2π radians, so the usual frequency we here, labelled ν is equal to $\omega/2\pi$. The factors of 2π are specific to sines and cosines,

but the point that the input to a function often looks like $kx - \omega t$ is very general. In that case, signals move at a speed of $\frac{\omega}{k}$. For *e.g.* string instruments, we can use Equation 10 to work out the frequency of a sound that an instrument will produce (given by ω) if we know the length of the instrument (which tells us k) and the speed at which signals move, since $\omega = vk$.

6. Summary and Discussion

For clarity, we picked a particular function to show how $f(x - vt)$ behaves. The function we picked has a peak, so we can watch the location of that peak to understand what's going on. The shape of the function doesn't change, but its position does as we change time. The larger v is, the faster the position moves as we change t . We would see the same general behavior with *any* function, where there's some underlying shape (*e.g.* what the function looks like at $t = 0$), and as we change time, the shape stays the same, but the position moves. We also saw that we can write functions to have dimensionless inputs, in which case they look like $f(kx - \omega t)$. When written like that, the wave speed is $v = \frac{\omega}{k}$.